Department of Physics

Advanced Laboratory Course: Particle physics

Search for $t\bar{t}$ resonances with ATLAS data

Practice of modern data analysis in high energy physics, using the ATLAS detector.

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Abstract

Particle physics is at a point of a very statistical and gradual unrevealing of the future of science. Most of the processes that are of interest in our discipline occur within time and length scales that are practically impossible to record or observe. The strong and electroweak interactions that we are trying to understand can only be probed by the rigorous analysis of the final and initial states of the particles involved in these interactions. The strategy scientists decided to employ in order to study these fundamental interactions has evolved ever so slightly since the 50's, when CERN was founded. The true evolution occurred in our detector and computational capacity. Today, in the Large Hadron Collider (LHC), we are colliding protons with 13 TeV center-of-mass energies at 4×10^8 times per second. A high center-of-mass energy and a frequent collision rate unlocks the possibility of analyzing rarer processes or those that are suppressed at lower energies. This increased capacity comes with a caviat; the perfect physics breeding ground we have at our collision point results in huge amounts of physical processes to occur per collision and accordingly, massive amount of combination data (including background and unresolvable events) is registered for each collision event.

Our purpose built detectors manage to record the final states of the fundamental particle interactions very well; however, the flipside of the coin, the massive data that we collect needs to be analyzed with many layers of similarly purposefully designed hardware systems, software systems and algorithms. Usually, we are interested in figuring out the details of a specific decay process. The detector measurement for any extremely rare hard scattering process in LHC is buried under statistical uncertainty of every other process, such as QCD multi-jet events, therefore we need to come up with a multi-layered strategy to systematically select only the signal for the process we want to study. In this laboratory course, we are going to conduct a comprehensive data analysis where we will look for a theoretical extension to the Standard Model "Z'" particle decaying into a "$t\bar{t}$" pair, utilizing the data from the ATLAS detector at LHC.

Contents

1 The ATLAS Detector **2**

2 Production and Decay of $t\bar{t}$ **4**

3 Probing Beyond the Standard Model: $Z' \to t\bar{t}$ **7**

4 Analysis Strategy **8**

5 Selection Cuts **8**

6 Fundamental distributions **9**

7 Discriminant **11**
7.1 Data-Monte Carlo Agreement Validation 13

8 Statistical Analysis **15**
8.1 Incorporating Systematic Uncertainties 15
8.2 Exclusion Limits at 95% Confidence Level 15

Conclusions **17**

1 The ATLAS Detector

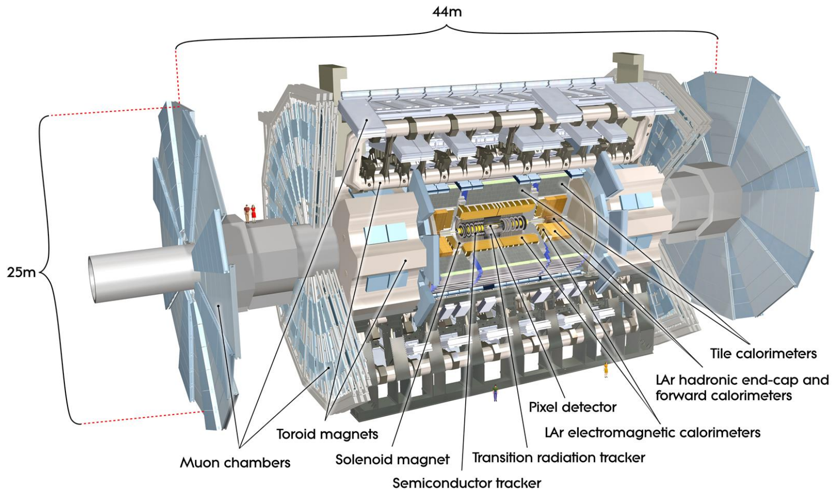

Figure 1: Cutout view of the ATLAS detector [**?**].

The ATLAS detector is a particle detector with a forward-backward symmetric cylindrical geometry and near 4π coverage in solid angle. Its sub-detectors are arranged in concentric layers around the interaction point. Each layer is specialized to detect or measure particular properties of the particles produced in proton-proton collisions. Following is a description of the major components, moving from the innermost to the outermost layers:

- **Inner Detector (ID)**:
  This subsystem is responsible for tracking charged particles and reconstructing their trajectories. It is immersed in a 2 T solenoidal magnetic field to allow momentum measurement via curvature. It consists of:
  - *Pixel Detector*: Provides the highest granularity and is located closest to the beam pipe. It enables precise vertex reconstruction, crucial for identifying primary and secondary vertices (e.g., from b-quark decays). Operates within $|\eta| < 2.5$ range.
  - *Semiconductor Tracker (SCT)*: Composed of silicon microstrip detectors, it offers precision tracking at intermediate radii. Also operates within $|\eta| < 2.5$ range.
  - *Transition Radiation Tracker (TRT)*: A straw tube tracker providing continuous tracking and electron identification via transition radiation. Operates within $|\eta| < 2$ range.

- **Calorimeters**:
  These measure the energy of particles by absorbing them and sampling the resulting showers. They are divided into:
  - *Electromagnetic Calorimeter (ECAL)*: Built from lead and liquid argon (LAr), it measures energies of electrons and photons with high resolution. Provides coverage up to $|\eta| < 3.2$.
  - *Hadronic Calorimeter (HCAL)*: Measures energy of hadrons (e.g. pions, kaons). The barrel region uses scintillating tiles and iron absorbers, while endcap and forward regions use copper/tungsten and LAr. Provides coverage up to $|\eta| < 4.9$.
- **Muon Spectrometer (MS)**:
  The outermost system, optimized to detect and measure muons. It uses large air-core toroidal magnets and multiple layers of tracking chambers (MDT, CSC) and trigger chambers (RPC, TGC). It provides precise standalone muon momentum measurements. Operates effectively within $|\eta| < 2.7$.
- **Magnet Systems**:
  ATLAS uses two types of magnets:
  - *Solenoid Magnet*: Surrounds the Inner Detector, producing a 2 T magnetic field for charged particle tracking. Covers the same region as the ID ($|\eta| < 2.5$).
  - *Toroidal Magnets*: Eight large air-core coils in the barrel and two endcap toroids provide magnetic fields for the Muon Spectrometer. Coverage up to $|\eta| < 2.7$.
- **Trigger and Data Acquisition System (TDAQ)**:
  This system selects potentially interesting events from the millions of collisions per second. It consists of a Level-1 hardware trigger and a high-level software trigger system.

A hypothetical Z' boson decaying into a top-antitop quark pair ($Z' \to t\bar{t}$) would lead to different final states rich in leptons, jets, b-jets, and missing transverse energy depending on the decay mode of the subsequent top quark pair ($t\bar{t}$). Charged leptons (electrons or muons) emerging from the decay of the W bosons would first be tracked in the Inner Detector and then deposit energy in the Electromagnetic Calorimeter; muons also traverse the calorimeters and are measured with high precision in the Muon Spectrometer. Jets arising from the hadronization of quarks (including b-quarks) would be reconstructed from energy clusters in the Hadronic Calorimeter and identified as b-jets using precise vertexing in the ID. Neutrinos would pass the detector completely undetected but this would cause an imbalance in transverse momentum (p_t) and reflect on the missing energy ($E_{\mathrm{T}}^{\mathrm{miss}}$), calculated from all other reconstructed objects. Together, these detector elements enable us to conduct the full reconstruction and identification of the $Z' \to t\bar{t}$ decay topology, allowing potential new physics signatures to be distinguished from Standard Model backgrounds.

2 Production and Decay of $t\bar{t}$

Top quark pairs ($t\bar{t}$) are dominantly produced through strong interaction in high-energy proton-proton collisions in processes such as gluon fusion (Figure 2). Once produced, the top and anti-top quarks decay almost exclusively via the weak interaction into a W boson and a bottom-type quark.

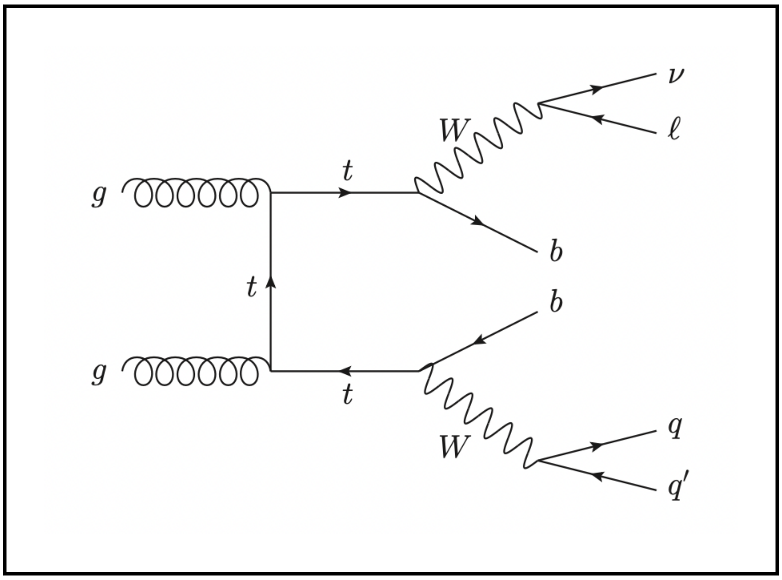

Figure 2: Feynman diagram of gluon fusion-$t\bar{t}$ production which later decays into lepton+jets channel.

The decay channels of the $t\bar{t}$ pair events are thus governed by the subsequent decays of the (two) W bosons, each of which can decay either hadronically (into a pair of quarks) or leptonically (into a lepton and neutrino). Decays of the W boson into a different generation quark-antiquark is possible, but it is CKM supressed and these cases do not constitute an additional decay branch. In experimental analyses, each final state has its strengths and weaknesses.

Depending on the decay modes of the two W bosons, $t\bar{t}$ events are categorized into three primary final states:

- **All-Hadronic Channel** ($t\bar{t} \to b\bar{b}q\bar{q}'q''\bar{q}'''$) In this channel, both W bosons decay into quark-antiquark pairs, resulting in a final state composed solely of jets. Alongside the two b-jets from the top quark decays, the event contains four additional light-flavor jets from the W bosons. Although this channel has the largest branching ratio (about 44%), it suffers from significant QCD multijet background, making signal discrimination challenging.

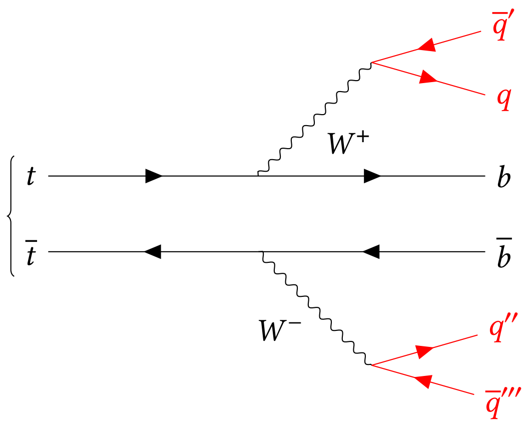

Figure 3: Feynman diagram for the Fully Hadronic decay channel.

- **Lepton+Jets Channel** ($t\bar{t} \to b\bar{b}\nu\bar{l}q\bar{q}'$) Here, one W decays leptonically and the other hadronically. The final state includes one charged lepton (electron or muon), missing transverse energy from the undetected neutrino, two light-flavor jets, and two b-jets. This channel offers a balance between background suppression and signal yield, with a branching fraction of about 44% (all lepton flavors). The presence of a high-p_t (transverse momentum) lepton and E_T^{miss} (missing transverse energy) makes it particularly suitable for triggering and analysis.

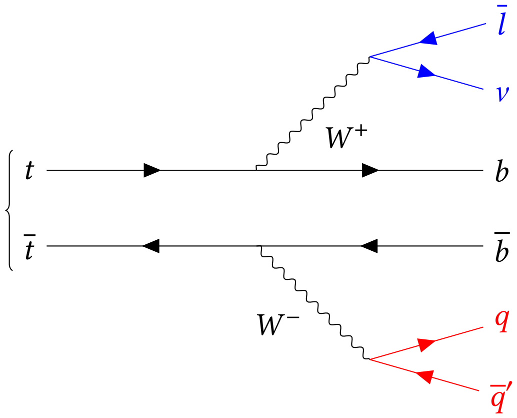

Figure 4: Feynman diagram for the Lepton + Jets decay channel.

- **Dileptonic Channel** ($t\bar{t} \rightarrow b\bar{b}\nu\bar{l}l'\bar{\nu}'$) In the dilepton channel, both W bosons decay leptonically, leading to two oppositely charged leptons (electrons or muons), missing transverse energy from the neutrinos, and two b-jets. This channel is the cleanest in terms of background (particularly from QCD), but it has the lowest branching fraction (about 11%). The reconstruction of the event is more difficult due to the presence of two neutrinos, which escape detection.

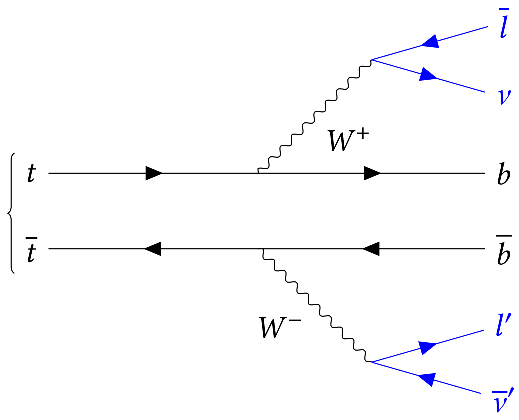

Figure 5: Feynman diagram for the Dileptonic decay channel:

3 Probing Beyond the Standard Model: $Z' \rightarrow t\bar{t}$

The analysis of the $Z' \rightarrow t\bar{t}$ process is a direct probe of physics beyond the Standard Model (BSM) in the search for a heavy BSM boson. Many BSM theories predict the existence of additional neutral gauge bosons (Z') or other heavy states that preferentially decay into top-quark pairs due to their large Yukawa coupling. The kinematic properties of high-mass resonance $t\bar{t}$ production in proton-proton collisions produce distinct signatures in the ATLAS detector. Some examples are large transverse momenta, boosted topologies, collimated jets from hadronic decays of boosted quarks and large missing transverse energy due to leptonic decays.

In this analysis we will leverage the reconstruction of invariant mass distributions of $m_{t\bar{t}}$, to search for a resonance bump above the falling Standard Model background. A statistically significant bump in the $m_{t\bar{t}}$ spectrum could indicate the presence of a new heavy particle. Limits on the production cross section times branching ratio of a hypothetical Z' boson are derived as a function of its mass. This enables constraints to be placed on specific BSM models or, in the case of a signal-like excess, motivates further investigation.

4 Analysis Strategy

This analysis searches for a spin-1 Z' resonance ($m_{Z'} = 0.5 - 3\,\text{TeV}$) decaying to $t\bar{t}$ pairs. The lepton+jets channel was selected for its optimal balance of branching fraction (~30%), manageable backgrounds, and compatibility with single-lepton triggers.

Data and Simulation:

- *Data:* pp collisions at $\sqrt{s} = 13\,\text{TeV}$ ($\mathcal{L} = 1\,\text{fb}^{-1}$) with single-lepton triggers ($E_T > 25\,\text{GeV}$ for e, $p_T > 25\,\text{GeV}$ for μ)
- *Backgrounds:* Simulated $t\bar{t}$, single-top, W/Z+jets, and diboson processes
- *Signal:* Simulated narrow-width Z' ($\Gamma/m \sim 1\%$) at benchmark masses (0.5, 0.7, 1.0, 1.5, 2.0, 3.0 TeV) with SM-like couplings

Reconstruction and Selection:

- *Objects:* Isolated leptons ($p_T > 25\,\text{GeV}$), anti-k_t jets ($R = 0.4$, $p_T > 25\,\text{GeV}$), b-tagging (70% WP), and $E_T^{\text{miss}} > 20\,\text{GeV}$
- *Event selection:* Exactly 1 lepton, ≥4 jets (2 b-tagged), $m_T(W) > 30\,\text{GeV}$
- *Top reconstruction:* Hadronic top (3 jets, 1 b-tagged) and leptonic top (lepton + E_T^{miss} + b-jet)

Discriminant: The reconstructed $t\bar{t}$ invariant mass $m_{t\bar{t}}$ serves as the final observable for resonance identification.

5 Selection Cuts

The event selection follows a sequential cut flow implemented in `runSelection.C`. The applied cuts are:

Step	Selection Criteria	Value
1	Exactly one lepton	`lep_n = 1`
2	Lepton transverse momentum	$p_T > 50$ GeV
3	Lepton pseudorapidity	$\|\eta\| < 2.5$
4	Jet multiplicity	$3 \leq N_{\text{jets}} \leq 5$
5	Leading jet p_T threshold	≥90 GeV
6	b-tagging requirement (MV2c10)	> 0.83
7	Missing transverse energy	$E_T^{\text{miss}} > 40$ GeV

Figure 6 shows the cumulative efficiency after these cuts. Background processes (especially $t\bar{t}$) experience significant suppression, though data reduction suggests no strong signal presence.

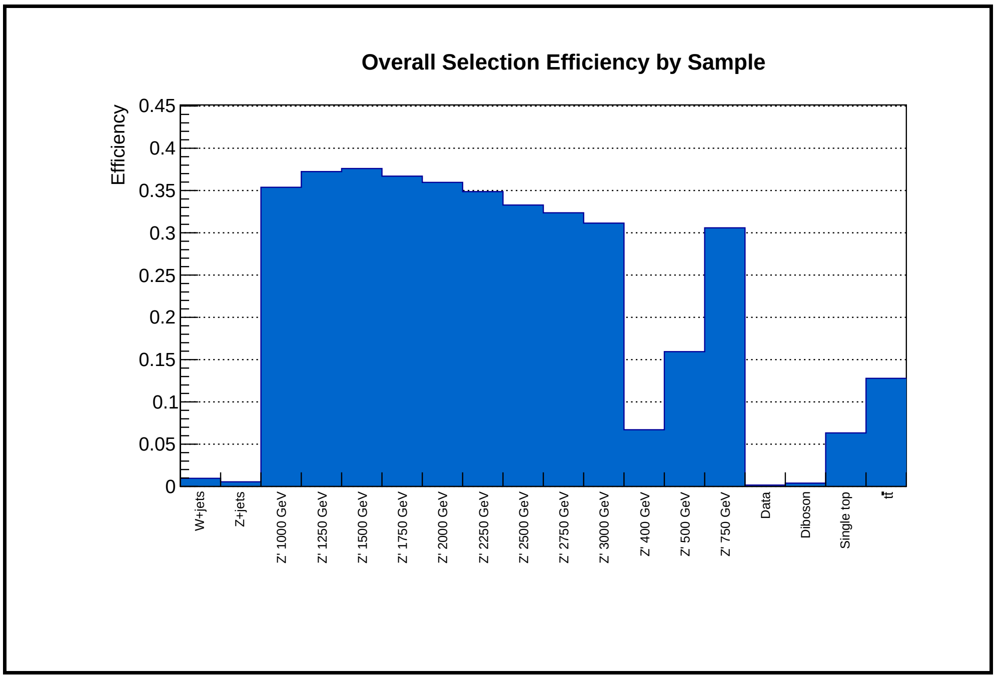

Figure 6: Overall efficiency result of the cuts on all samples.

6 Fundamental distributions

The dominant background after event selection is strong $t\bar{t}$ production due to its identical final state to the signal, resulting in high selection efficiency. Kinematic distributions for the $t\bar{t}$ sample were examined, including:

- Lepton properties: p_T, η, ϕ, E
- Jet properties: p_T, η, ϕ, E for all jets and leading jets
- Multiplicities: Number of jets and b-tagged jets
- Missing transverse momentum ($p_{T,\text{miss}}$)

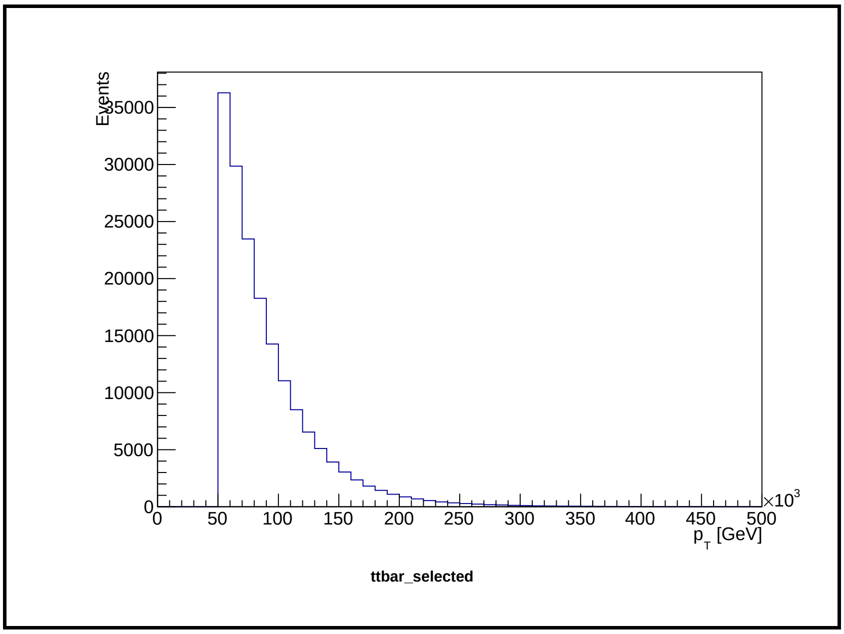

Figure 7: Lepton transverse momentum.

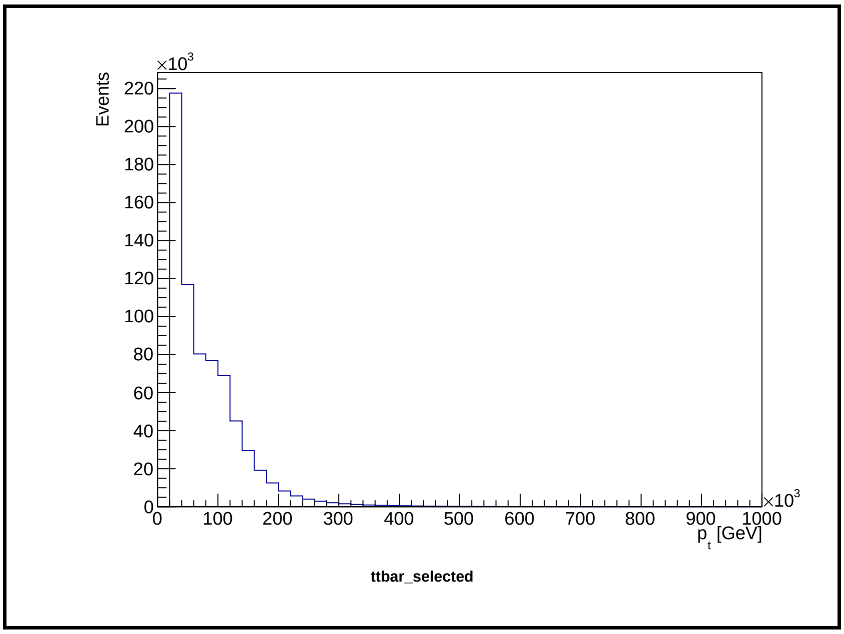

Figure 8: Jet transverse momentum.

Both lepton and jet p_T exhibit exponential decays, while Jet and b-jet multiplicities show similar distributions.

All distributions align with SM expectations, confirming background modeling validity. Consequently, these observables alone cannot serve as the final discriminant in a resonance search.

7 Discriminant

To enhance the signal-to-background separation in the search for $t\bar{t}$ resonances, multiple kinematic discriminants were investigated. These discriminants leverage the full event topology by combining four-vectors of reconstructed objects. The primary candidates considered were:

- Missing transverse momentum ($p_{\mathrm{T,miss}}$)
- Azimuthal angle separation between $p_{\mathrm{T,miss}}$ and the lepton ($\Delta\phi(\ell, p_{\mathrm{T,miss}})$)
- Invariant mass of the three highest-p_T jets ($m(3j)$)
- Invariant mass of the reconstructed $t\bar{t}$ system ($m_{t\bar{t}}$), formed by:

$$m_{t\bar{t}} = m(4j, \ell, \nu) = m(p_{t_{\mathrm{had}}} + p_{t_{\mathrm{lep}}})$$

- Pseudorapidity of the full $t\bar{t}$ system ($\eta(4j, \ell, \nu)$)

Distributions of these variables were compared between simulated $t\bar{t}$ background and Z' signal events (e.g., $m_{Z'} = 1000\,\mathrm{GeV}$). While $\Delta\phi$, $m(3j)$, and $\eta(4j, \ell, \nu)$ exhibited similar shapes for both samples (with larger fluctuations in the Z' sample due to limited statistics), the $p_{\mathrm{T,miss}}$ distribution showed an exponential decrease for background but no clear functional form for signal. Crucially, $m_{t\bar{t}}$ displayed fundamentally different behavior:

- For background, a broad peak near 500 GeV (higher than the expected $2m_t \approx$ 346 GeV, suggesting MC modeling limitations)
- For Z' signal, a sharp resonance at the generated mass (e.g., ~1100 GeV for $m_{Z'} =$ 1000 GeV)

The reconstructed $m_{t\bar{t}}$ demonstrated superior discrimination power due to its direct physical interpretation: it should peak at the resonance mass for a $Z' \to t\bar{t}$ signal while yielding a smoothly falling distribution for SM backgrounds (Figure 9). This clear signature motivated its selection as the final discriminant for statistical analysis in both studies. Other variables served auxiliary roles in validation and systematic uncertainty studies but lacked comparable sensitivity to narrow resonances.

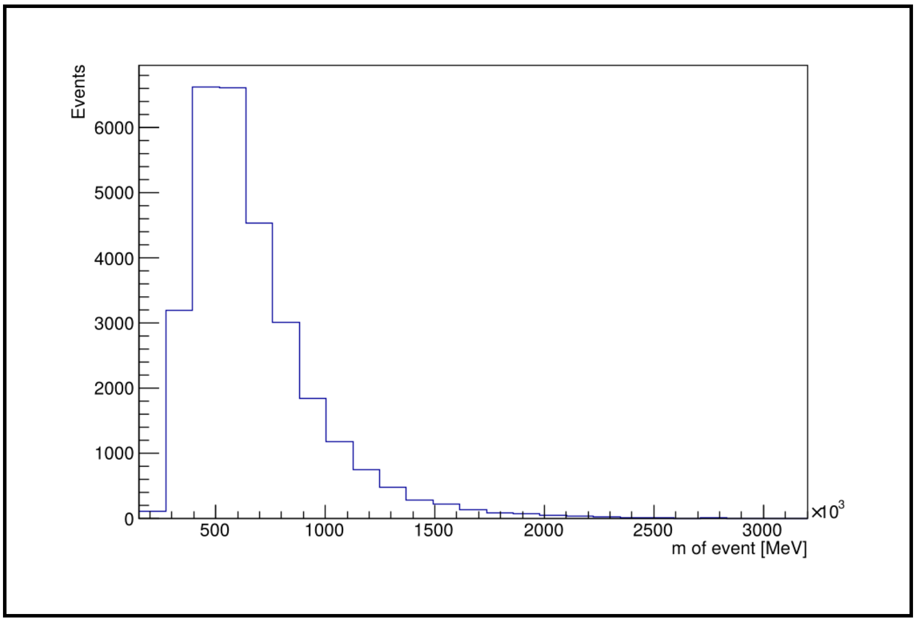

Figure 9: Reconstructed invariant mass of the four jets with the highest transverse momentum, the lepton, and missing transverse momentum ($m_{t\bar{t}}$) for SM background.

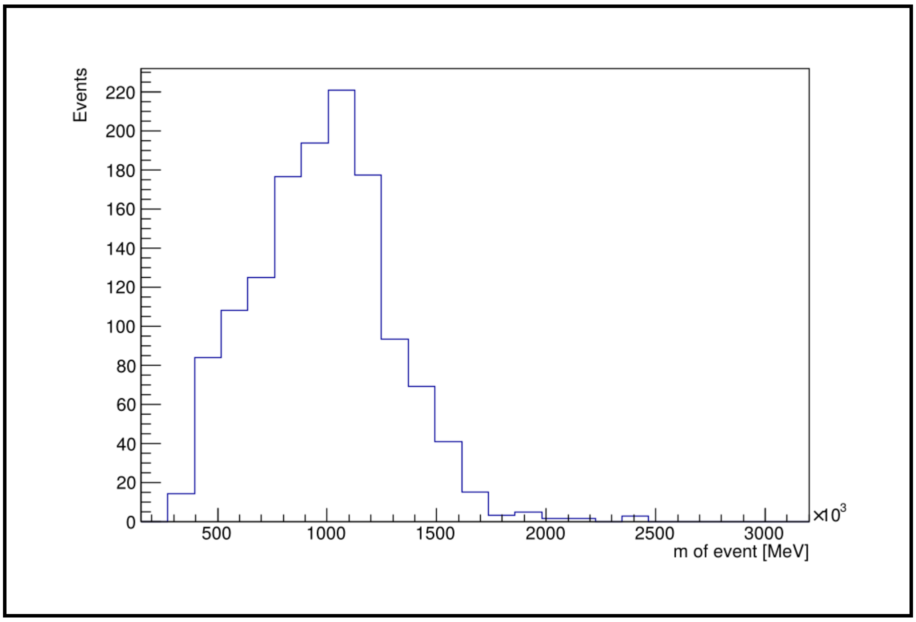

Figure 10: Reconstructed invariant mass of the four jets with the highest transverse momentum, the lepton, and missing transverse momentum for Z' signal ($m_{Z'}$ = 1000 GeV).

7.1 Data-Monte Carlo Agreement Validation

Prior to conducting the resonance search, rigorous validation of the agreement between collision data and Monte Carlo (MC) simulations is essential. This verification ensures the reliability of background modeling in control regions and establishes confidence in the statistical interpretation framework. The validation procedure comprises two complementary approaches:

1. **Event Yield Comparison:** The expected event yields for Standard Model (SM) processes are calculated as:

   $$N_{\text{expected}} = \mathcal{L} \cdot \sigma \cdot (A \cdot \epsilon)$$

   where $\mathcal{L}$ denotes the integrated luminosity, σ the process cross-section, and $(A \cdot \epsilon)$ the selection efficiency. For the dataset, an excess of observed events compared to the expected number is observed. This discrepancy persists across kinematic distributions but lacks a resonant mass peak, suggesting systematic mismodeling rather than new physics.

2. **Kinematic Distribution Analysis:** Stacked histograms compare data and MC distributions for key observables, normalized using per-event weights:

   $$w = \frac{\text{generator weight} \times \mathcal{L} \times \sigma}{\sum \text{generator weights}}.$$

   Lower plots display the data/MC ratio with statistical uncertainties. Results show:

   - *Lepton p_T*: Agreement within ±20% across p_T bins, with minor deviations > 200 GeV consistent with statistical fluctuations.
   - *Reconstructed $m_{t\bar{t}}$*: Smoothly falling spectrum with discrepancies between Data and MC reconstruction.

Indeed for the mass distribution can be spotted some deviations (e.g. around 635 GeV and 900 GeV) that suggest for a shift in the Data distribution more to the right. The observed yield excess is attributed to limitations in MC simulations (e.g., imperfect b-tagging efficiency modeling or parton shower tuning) rather than a signal. This validation justifies proceeding to statistical analysis of the final discriminant.

We therefore conclude that the background modeling is reliable in the control regions, and proceed to inspect the final mtt̄ spectrum for evidence of a signal. Just by looking at it, we do not see any indication of a signal.

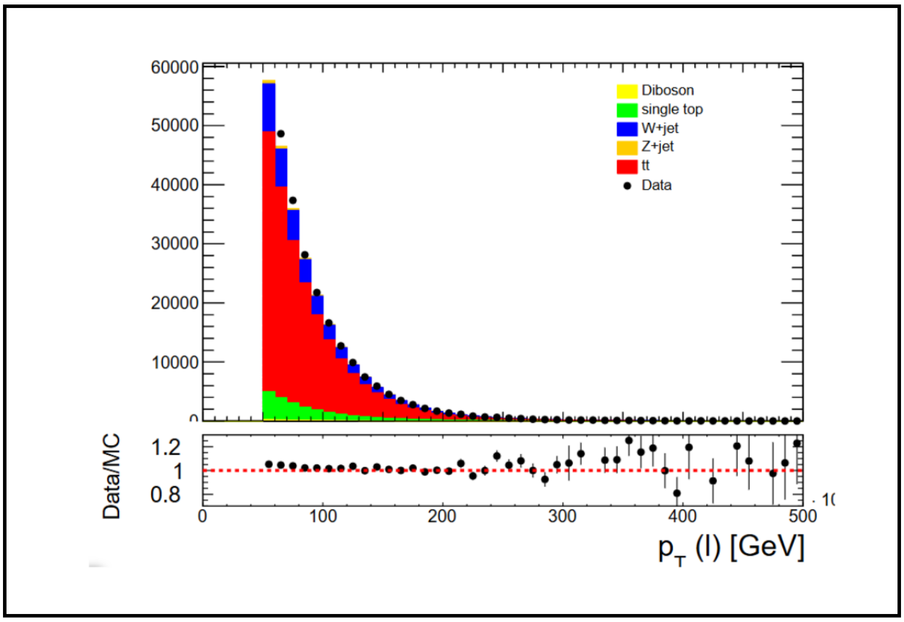

Figure 11: Stacked comparisons of lepton p_T and between data and MC. The lower panel shows the data/MC ratio with statistical uncertainties.

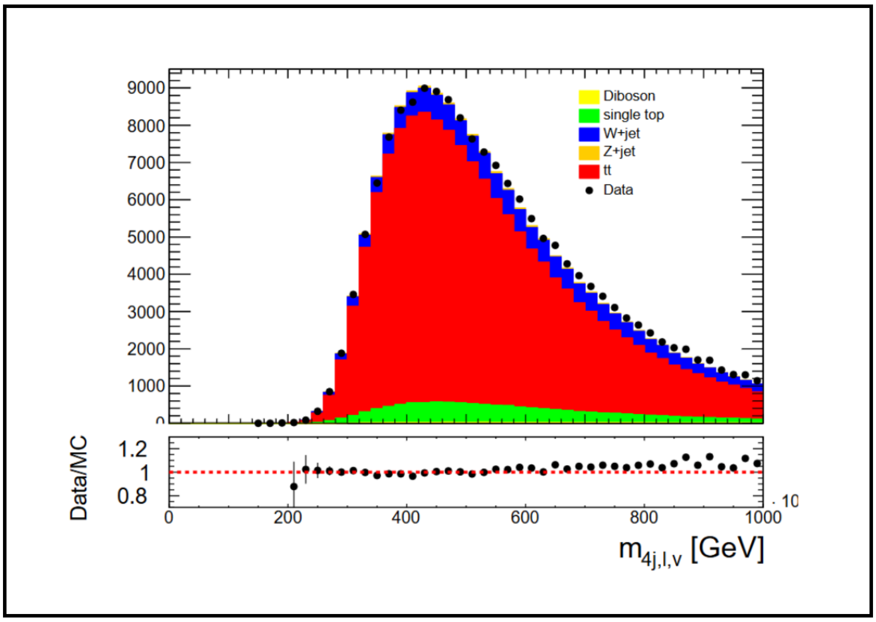

Figure 12: Stacked comparisons of reconstructed $m_{t\bar{t}}$ between data and MC.

8 Statistical Analysis

The agreement between observed data and simulated Standard Model (SM) background predictions was quantified using the χ^2 test statistic. For the final discriminant variable (reconstructed $t\bar{t}$ invariant mass $m_{t\bar{t}}$), the χ^2 value is computed as:

$$\chi^2 = \sum_{i=1}^{N_{\text{bins}}} \frac{(N^i_{\text{data}} - N^i_{\text{MC}})^2}{N^i_{\text{MC}}}$$

where N^i_{data} and N^i_{MC} represent event counts in bin i for data and simulated background, respectively. The corresponding p-value is derived from the χ^2 distribution with $k = N_{\text{bins}} - 1$ degrees of freedom.

A p-value below 2.7×10^{-3} (3σ) is considered an *evidence*, while a p-value below 5.7×10^{-7} (5σ) is considered an *observation.*

The calculated p-value of 1.34×10^{-2} indicates a deviation of approximately 2σ. This value is above the conventional threshold and thus does not constitute significant evidence for a signal above the background. Furthermore, this statistical analysis does not incorporate systematic uncertainties, which, as discussed in the following section, may substantially influence the final result.

8.1 Incorporating Systematic Uncertainties

To quantify systematic uncertainties, the χ^2 test was recomputed with a uniform 14% per-bin uncertainty. This comprehensive uncertainty combines:

- 4% integrated luminosity uncertainty
- 10% experimental uncertainties (from b-tagging efficiency, lepton identification, etc.)

The modified bin variance accounts for these systematics through:

$$\sigma^2_{\text{tot}} = \sigma^2_{\text{stat}} + (0.14 \times N_{\text{bkg}})^2$$

where σ_{stat} is the statistical uncertainty and N_{bkg} is the background yield per bin.

This leads to a p-value equal to 0.998: no Z' boson was so discovered.

8.2 Exclusion Limits at 95% Confidence Level

In the absence of statistically significant evidence for a signal, we derive 95% confidence level (CL) upper limits on the production cross section times branching ratio

$\sigma(pp \to Z') \times \mathcal{B}(Z' \to t\bar{t})$. For each hypothesized resonance mass $m_{Z'}$, we introduce a signal template superimposed on the SM background and systematically vary its normalization. The normalization is adjusted until the χ^2 probability comparing data to the combined background-plus-signal model falls below 0.05 (i.e., $1-0.95$). The resulting exclusion curve ("observed limit") is plotted against theoretical predictions, as shown in Figure 13.

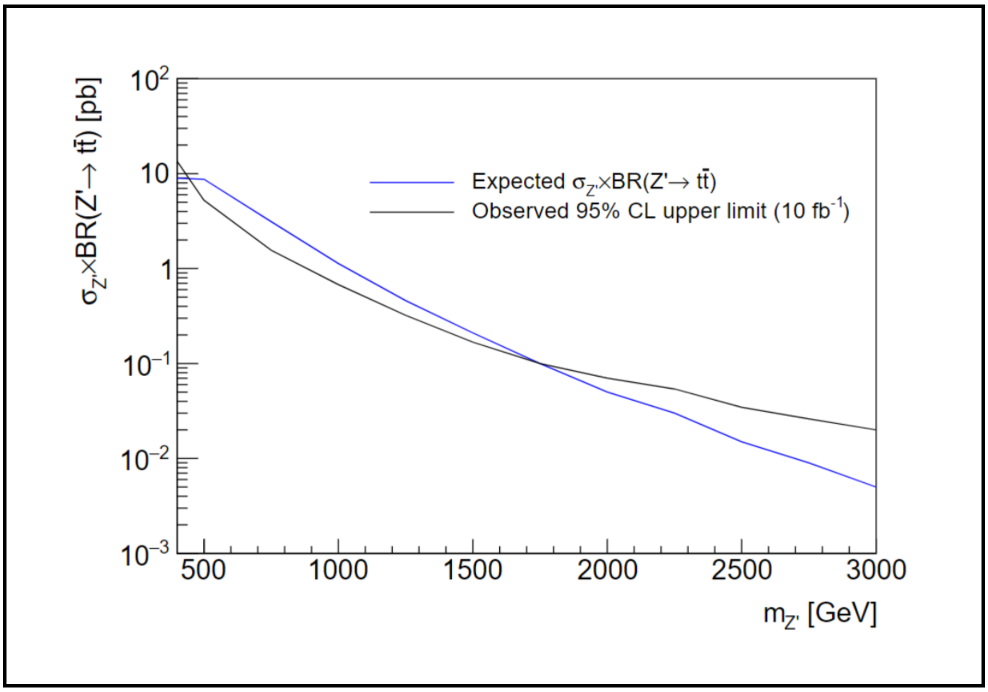

Figure 13: Theoretical limits and 95% CL exclusion limits on the production cross section for different Z' boson masses.

- **Expected limit (blue curve)**: The median upper limit under the background-only hypothesis (no true Z' signal), quantifying the analysis sensitivity to a narrow resonance of mass $m_{Z'}$ given statistical fluctuations of SM backgrounds.
- **Observed limit (black curve)**: Actual upper limits derived from collision data. Fluctuations in data relative to expectations shift this curve above/below the expected limit.

The vertical axis, shown on a logarithmic scale, represents the fraction of the Z' production cross section corresponding to the $Z' \to t\bar{t}$ decay channel, expressed in pb. The horizontal axis denotes the hypothesized Z' boson mass, scanned from 500 GeV to 3 TeV.

The expected limit (blue curve) represents the median expected 95% confidence level (CL) upper limit under the background-only hypothesis, where no Z' signal is present. This curve characterizes the intrinsic sensitivity of the analysis to a narrow resonance of a given mass $m_{Z'}$. The observed limit (black curve) shows the actual 95% CL upper limit derived from the experimental data.

Statistical fluctuations in the data cause the observed limit to deviate from the expected median. An upward fluctuation of the signal-like background results in an observed limit less stringent than expected (i.e., above the blue curve), while a downward fluctuation yields a more stringent limit (i.e., below the blue curve).

This plot provides a direct means of evaluating theoretical models. For example, if any model that predicts values of $\sigma \times \mathcal{B}$ exceeding the black "observed" curve, then it is excluded at the 95% confidence level. As an example, consider a benchmark Sequential Standard Model Z' with a mass of $m_{Z'} = 2$ TeV , which predicts

$$\sigma \times \mathcal{B} \approx 0.2\,\mathrm{pb}$$

If the observed upper limit at 1 TeV is 0.1 pb, this specific model prediction would be excluded by the analysis.

Conclusions

In this analysis, a search for a narrow $Z' \to t\bar{t}$ resonance was performed using $1\,\mathrm{fb}^{-1}$ of $\sqrt{s} = 13$ TeV data from the ATLAS detector. The reconstructed $t\bar{t}$ invariant mass $m_{t\bar{t}}$ was selected as the final discriminant due to its superior separation power between signal and background. After applying optimized event selection criteria, a binned χ^2 test of the $m_{t\bar{t}}$ distribution was applied to evaluate the agreement between background and the observed data. That resulted in a p-value of 1.34×10^{-2} and thus does not provide evidence for a signal. However, the contribution of systematic uncertainties was not taken into account. After including their contribution, a p-value of 0.998 was obtained, showing no significant deviation from Standard Model expectations.

As no significant excess above the Standard Model background prediction was observed, exclusion limits at the 95% confidence level were derived on the Z' production cross section multiplied by its branching fraction to top-quark pairs, $\sigma(pp \to Z') \times \mathcal{B}(Z' \to t\bar{t})$. These limits were presented as a function of the hypothesized Z' mass, showing both the theoretical prediction and the 95% CL exclusion limits on the production cross-section. It was found that, for Z' masses up to several TeV, production rates above a few picobarns are excluded.